Exp 27 MATLAB CODE

```
clc;
clear;
close all;

% Parameters
N = 10;
data_bits = 10;
chips_per_bit = 2^N - 1;

% Generate Random Data Bits
data = randi([0 1],1,data_bits);

% Generate PN Sequence using LFSR
pn = zeros(1,chips_per_bit);
register = ones(1,N);

for i = 1:chips_per_bit
   pn(i) = register(end);
   feedback = xor(register(10),register(3));
   register = [feedback register(1:end-1)];
end

% Convert Data Bits to Bipolar Form
bipolar_data = 2*data - 1;

% Spread Data using PN Sequence
pn_bipolar = 2*pn - 1;
```

```
spread_signal = zeros(1,data_bits*chips_per_bit);

for i = 1:data_bits
   spread_signal((i-1)*chips_per_bit+1:i*chips_per_bit) = ...
      bipolar_data(i)*pn_bipolar;
end

% Processing Gain
processing_gain = chips_per_bit;
processing_gain_dB = 10*log10(processing_gain);

% Display Results
disp('Input Data Bits:');
disp(data);

disp('Generated PN Sequence:');
disp(pn);

fprintf('Chips per Bit = %d\n',chips_per_bit);
fprintf('Processing Gain = %d\n',processing_gain);
fprintf('Processing Gain in dB = %.2f dB\n',processing_gain_dB);

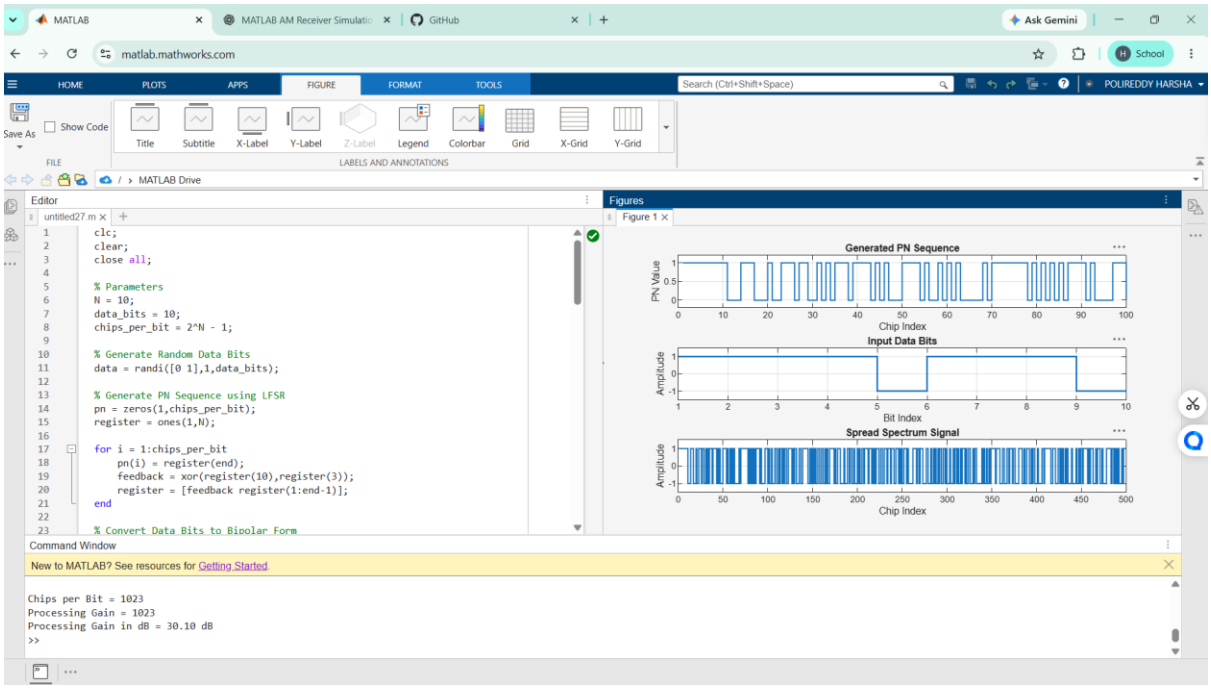
% Plot PN Sequence
figure;

subplot(3,1,1);
stairs(pn(1:100),'LineWidth',1.2);
title('Generated PN Sequence');
```

```
xlabel('Chip Index');
ylabel('PN Value');
ylim([-0.2 1.2]);
grid on;

% Plot Input Data
subplot(3,1,2);
stairs(bipolar_data,'LineWidth',1.5);
title('Input Data Bits');
xlabel('Bit Index');
ylabel('Amplitude');
ylim([-1.5 1.5]);
grid on;

% Plot Spread Signal
subplot(3,1,3);
stairs(spread_signal(1:500),'LineWidth',1.2);
title('Spread Spectrum Signal');
xlabel('Chip Index');
ylabel('Amplitude');
ylim([-1.5 1.5]);
grid on;
```


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LABELS AND ANNOTATIONS
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Editor
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clc;
clear;
close all;
% Parameters
N = 10;
data_bits = 10;
chips_per_bit = 2^N - 1;
% Generate Random Data Bits
data = randi([0 1],1,data_bits);
% Generate PN Sequence using LFSR
pn = zeros(1,chips_per_bit);
register = ones(1,N);
for i = 1:chips_per_bit
pn(i) = register(end);
feedback = xor(register(10),register(3));
register = [feedback register(1:end-1)];
end
% Convert Data Bits to Bipolar Form
Figures
Figure 1
Generated PN Sequence
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Chip Index
Command Window
New to MATLAB? See resources for Getting Started.
Chips per Bit = 1023
Processing Gain = 1023
Processing Gain in dB = 30.10 dB
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